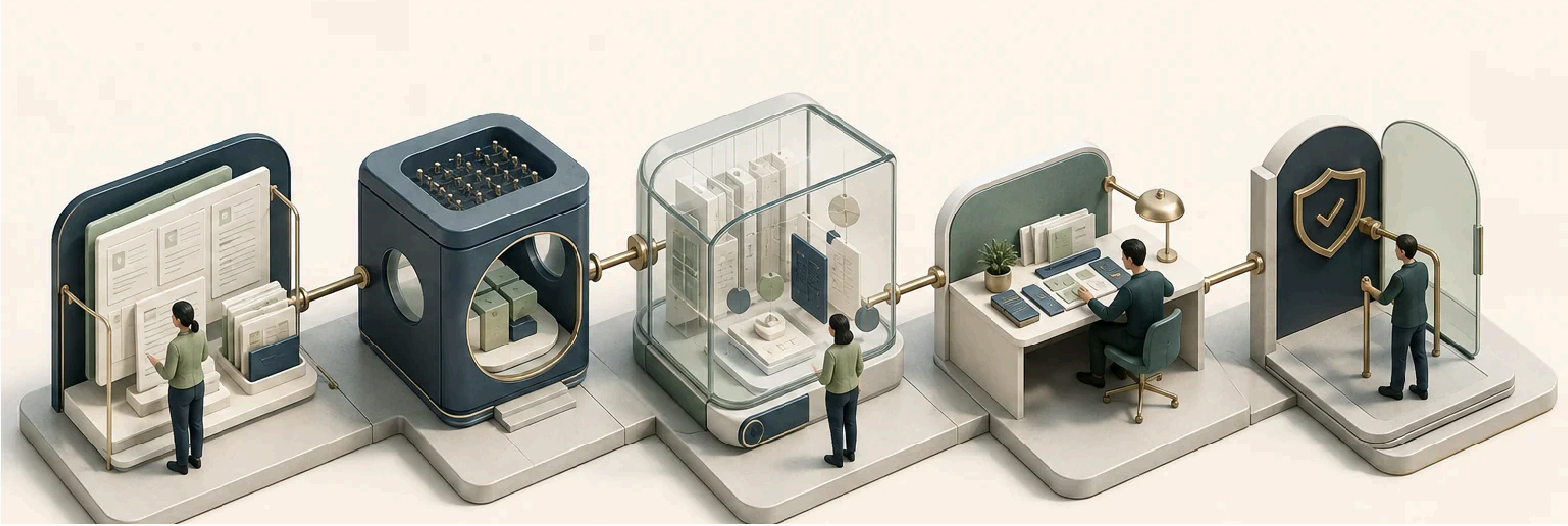
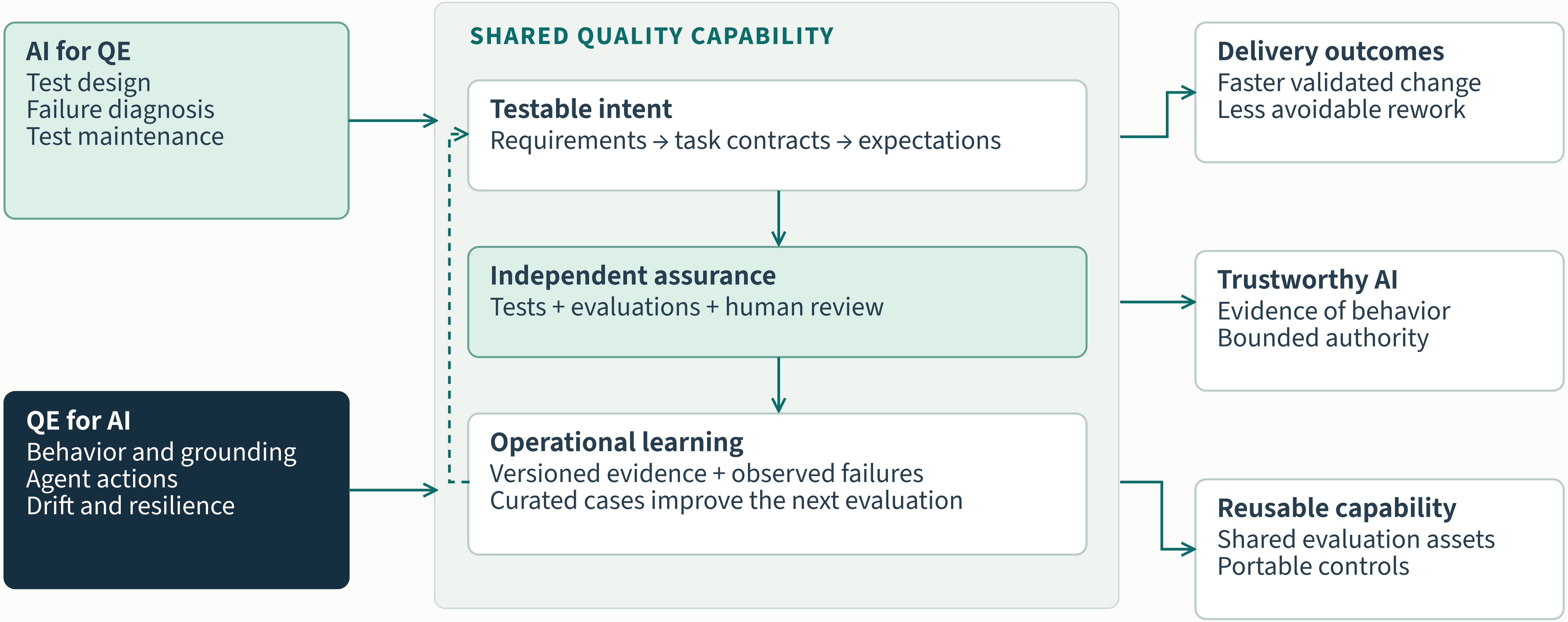
Quality engineering for the AI era

AI × quality engineering · Industry research



A shared quality capability

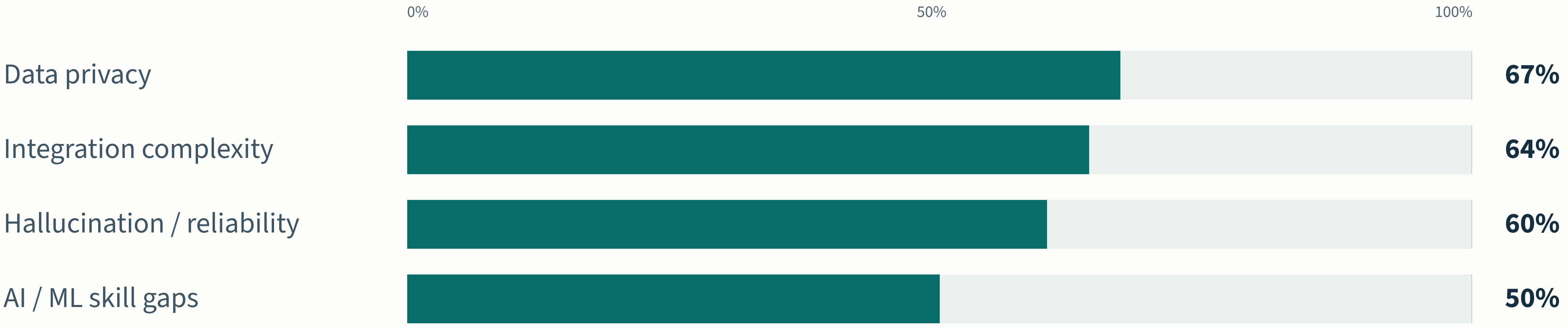


Strategic vision · outcomes are ambitions to validate, not reported bank results

The constraints are organizational

What constrains adoption

Share of respondents reporting each barrier · World Quality Report 2025–26



Overlapping responses, not shares of a whole. Survey: >2,000 executives; question-specific n unavailable in the public release. Source: Capgemini, Sogeti and OpenText, November 2025. [W01]

Task gains depend on the setting

CUI ET AL. / THREE FIELD EXPERIMENTS

+26.08%

completed tasks

Developer sample
4,867

Precision
SE 10.3 percentage points

Setting
Coding assistants in three firms

METR / EARLY-2025 TRIAL

+19%

completion time

Developer / task sample
16 / 246

Precision
95% CI: +2% to +39%

Setting
Experienced developers; familiar repos

The unit, task and user population determine what transfers.

Separate outcomes; do not pool these percentages or treat them as QE savings rates.

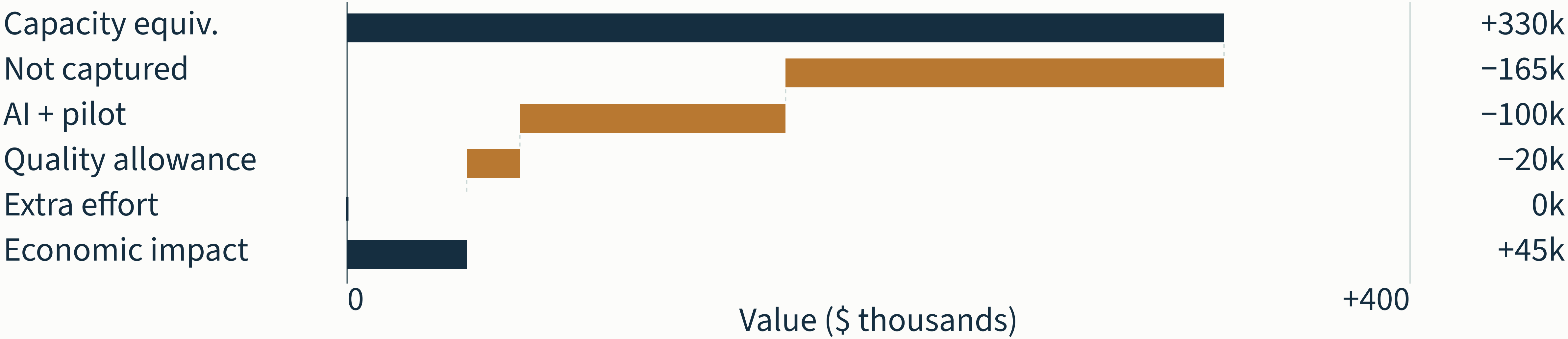
Capacity and realized value

$45k illustrative net economic impact

3.30% capacity released · 0.45% of addressable spend

Annual value bridge

$ thousands · per $10M addressable QA spend



Illustrative assumptions, not a forecast. Capacity is effort equivalent; cash savings need Finance-approved capture.

Adoption 50% · net task saving 20% · capacity captured 50%.

Activity share 55% · eligibility 60% · AI/pilot cost 1.0% · quality allowance 0.2%.

The industry direction and its limits

SOURCE / EVIDENCE TYPE	WHAT IT SAYS	STRATEGIC IMPLICATION
Gartner · forecast	90% of enterprise software engineers using AI code assistants by 2028.	Plan for widespread assisted work; tool use alone does not establish quality or savings.
McKinsey · survey	Nearly 300 leaders surveyed; 100 assessed impact. Top performers reported broader delivery improvements.	Invest in workflow and organizational change. These self-reported outcomes do not isolate causality.
DORA · observational research	AI interacts with the surrounding engineering system.	Improve feedback, testing and platform foundations alongside adoption.

Public sources, reviewed September 2026. Gartner's gated testing reports contribute public abstracts only; no proprietary vendor ranking is reproduced. [G01][G02][G03][M01][D01]

Industrial cases reveal reusable patterns

INDUSTRIAL EXAMPLE	OBSERVED APPROACH	CAPABILITY TO DEVELOP
Meta · TestGen-LLM / ACH	Generated tests face build, reliability and coverage checks; ACH adds fault-oriented validation.	An independent test-quality gate, with meaningful assertions.
Google · Auto-Diagnose	Failure summaries link likely causes to relevant log lines in the review workflow.	Diagnosis with inspectable evidence and reviewer feedback.
Uber / UT Austin · FlakyGuard	Targeted execution context supports flaky-test repair in an enterprise Go repository.	Repair validation that preserves the original test intent.

Company-specific cases demonstrate mechanisms. They do not establish a transferable bank savings rate or a product comparison. [E04][E05][E06][E07]

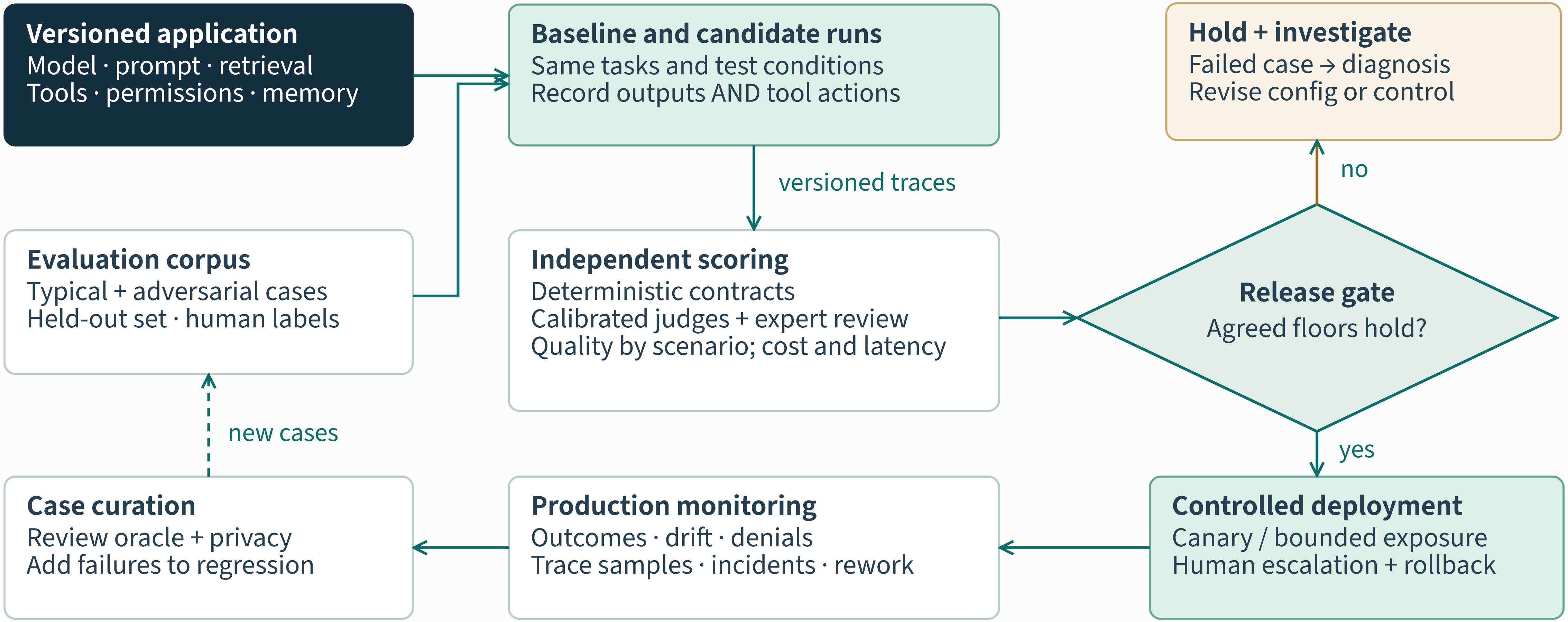
The first workflows need a clear quality check

	Advice / draft artifacts	Greater action authority
Easy to verify	**BOUNDED STARTING POINTS** Test drafts with an executable contract Failure summaries linked to logs	**CONTROLLED EXECUTION** Sandbox test execution Draft pull requests through scoped tools
Needs expert judgment	**BUILD THE EVALUATION CAPABILITY** Risk analysis and exploratory test ideas Expert review defines useful output	**SEPARATE AUTHORITY DECISION** Production changes and customer actions Require action-level evidence and recovery

What the findings change

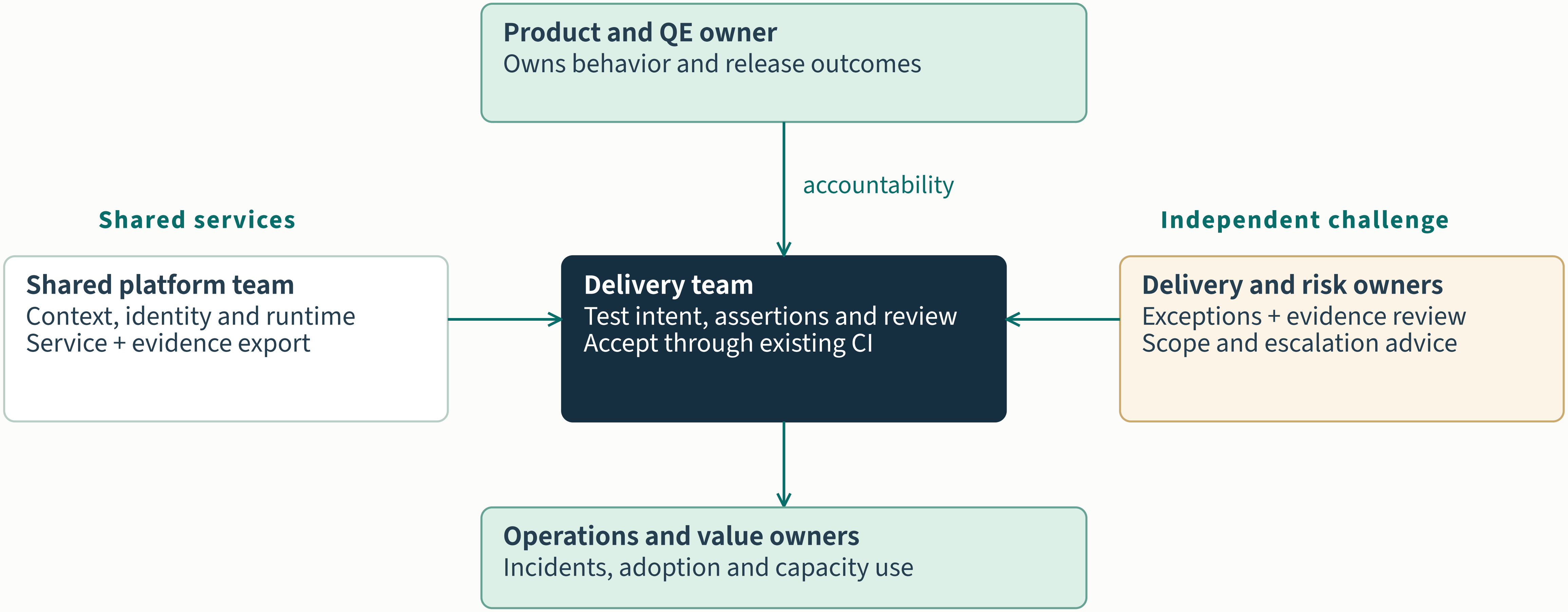
RESEARCH OBSERVATION	DESIGN RESPONSE	EVIDENCE TO RETAIN
DORA / verification effort Generation can shift work into review.	**Join effort with run telemetry** Measure net workflow effort	**Prep · review · correction** D02
Meta / filtered test generation Passing is not proof of added value.	**Independent test-quality gate** Protect approved assertions	**Stable runs · fault detection** E04 · E05
NIST / lifecycle assurance Assurance continues in production.	**Version cases; feed failures back** Re-evaluate configuration changes	**Case history · drift · release** A01 · A02
OWASP + OSFI / agent actions Tools expand the impact of failure.	**Enforce permissions outside the model** Test denied actions and recovery	**Identity · decision · approval** A03 · R01

Quality improves through feedback



Proposed application-level assurance · evaluate again when context, behavior or authority changes

Ownership across the quality lifecycle

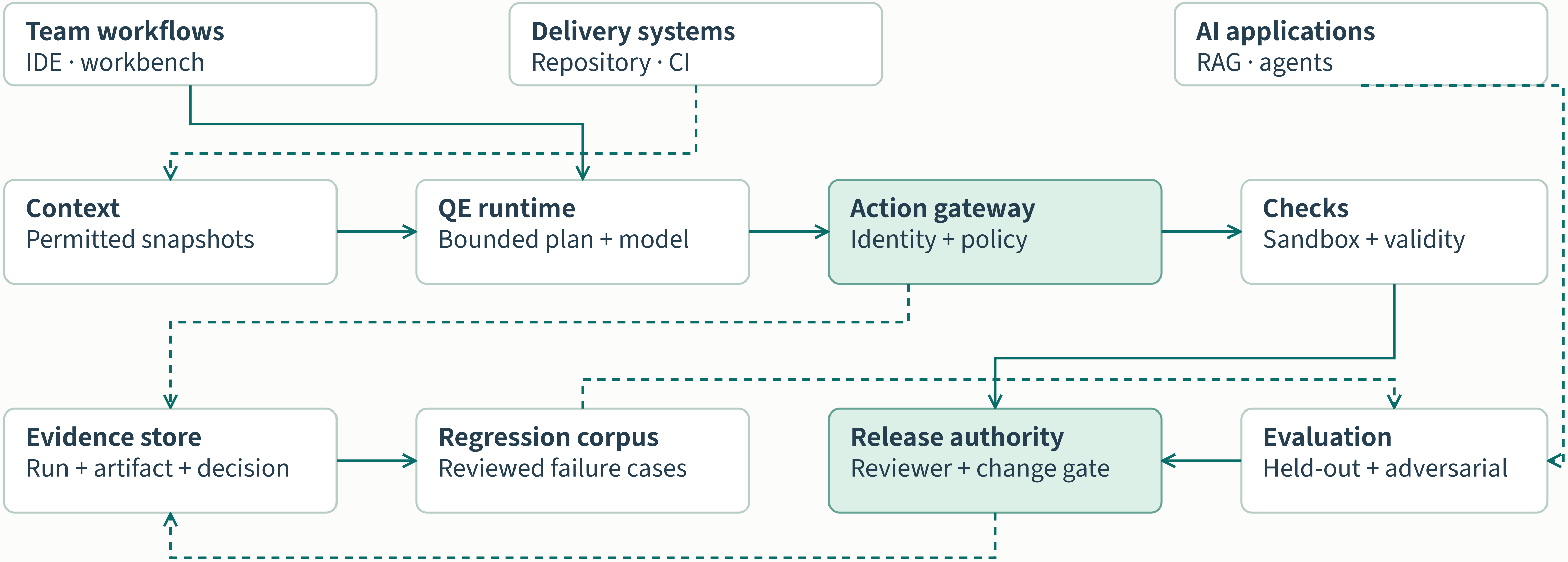


The quality role gains new disciplines

CAPABILITY	WHAT PRACTITIONERS NEED TO DO	OBSERVABLE LEARNING EVIDENCE
Test intent and oracles	Define expected behavior and challenge plausible but weak assertions.	A test catches a relevant fault and preserves the contract.
Evaluation and curation	Build representative cases; calibrate expert labels and model judges.	Evaluation results survive a held-out scenario review.
Agent assurance	Inspect tool authority, context boundaries and recovery behavior.	A denied-action exercise leaves no unauthorized side effect.
Workflow ownership	Measure review and correction work; coach teams through real tasks.	Task-level outcomes include the cost of verification.

Proposed capability agenda informed by research on verification work and lifecycle assurance; it is not a staffing reduction plan. [D02][G04][A01][E05]

The platform outlasts a model



Shared foundations and domain expertise

PRODUCT-SPECIFIC ASSETS

Payments
Transaction invariants
Failure and recovery cases

Customer service AI
Grounded response criteria
Escalation and tool scenarios

Engineering workflows
Repository test contracts
Review and rework baselines

REUSABLE ASSURANCE SERVICES

Dataset registry · Evaluation runners · Identity + gateway · Evidence store · Recovery tooling

REPLACEABLE MODEL AND TOOL ADAPTERS

Procure capabilities against local tests; preserve access to configurations, artifacts and evidence.

Evidence gates the next horizon

Gate: local quality + net effort

01

Trusted assistance

Draft tests + diagnose failures
Independent checks; human approval

Gate: repeatable outcomes across teams

02

Connected assurance

Shared evaluation and context services
Cross-team evidence and feedback

Gate: safe actions + tested resilience

03

Controlled agents

Multi-step work with bounded authority
Action-level controls and recovery

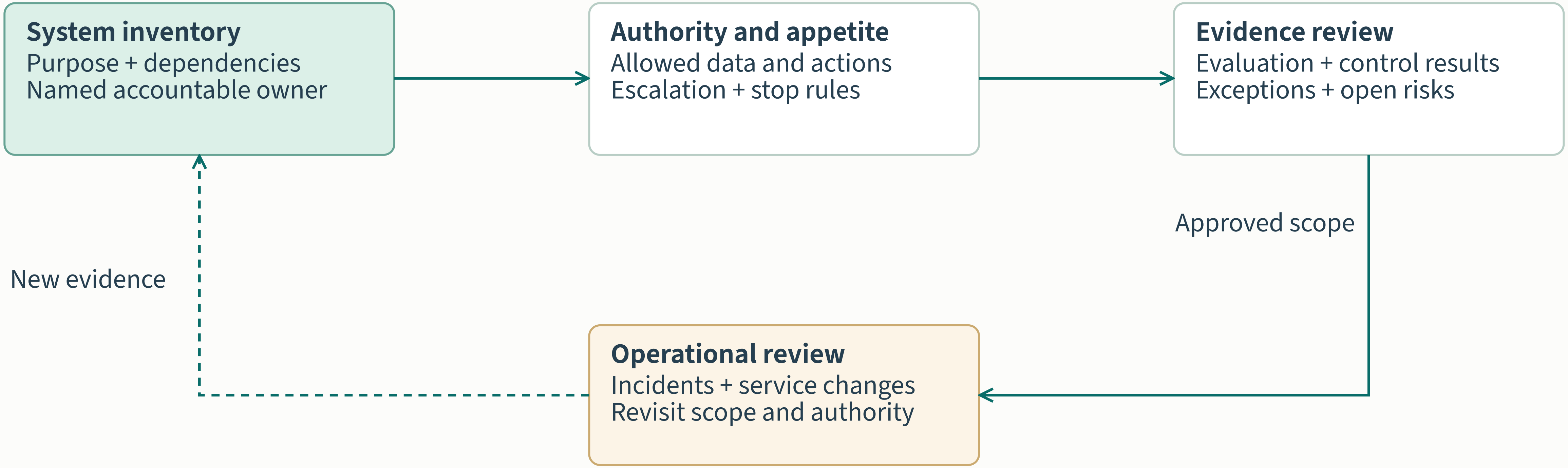
Proposed horizons, not a dated forecast or measured maturity score

The leadership evidence contract

STRATEGIC OUTCOME	EVIDENCE TO EXAMINE	LEADERSHIP DECISION
Delivery capacity	Net effort with review and rework	Redeploy verified capacity
Trustworthy outcomes	Fault detection and escaped defects	Hold the quality floor
Reusable capability	Results by team and task	Fund the shared service
Controlled autonomy	Denials and recovery tests	Set the next authority limit

Every readout includes task mix, sample, period, costs and uncertainty. No organizational results are claimed here.

Governance follows the system lifecycle



OSFI context: July 2026 bulletin = sound practices; revised E-23 takes effect 1 May 2027.

The strategic choices ahead

Ambition

Quality as a shared capability

Choose the delivery constraints and AI risks the organization needs to address.

Investment

Foundations and people

Develop evaluation skills, platform integration and accountable ownership.

Expansion

Value with evidence

Grow the capabilities that improve outcomes within their control boundary.

> Vision: AI contributes more work while the organization strengthens its ability to judge and govern that work.

Continue to the architecture ↗

Choose the shared assurance model

01

Local assistants
Fast individual adoption
Duplicated context and checks
Local learning stays fragmented

Use to learn tasks

02

Shared assurance
Portable tests and evidence
Domain-owned expectations
Common evaluation and controls

Recommended strategic center

03

Bounded agents
Multi-step delegated work
Stronger action authority
Higher recovery obligations

Earn through evidence

Sequencing choice: standardize the proof before expanding delegated authority.

Trade-off: shared services require ownership; agent breadth requires independently tested containment.

Research and implementation detail ↗

What changes in a payment-API workflow

CURRENT WORKING HYPOTHESIS · VALIDATE WITH THE BANK

Interpret requirement
Interpret intent

→

Write tests
Local prompts

→

Review change
Rebuild evidence

→

Release
Late evidence assembly

PROPOSED TARGET · NEGATIVE PAYMENT AMOUNTS MUST BE REJECTED

Domain contract
Owner-approved rule
Versioned test oracle

→

Bounded generation
Permitted context only
Sandbox tests

→

Independent proof
Original passes
Mutant fails
Reviewer decides

→

Evidence-led release
Existing release owner
Failure → new test case

Shared: identity, evaluation runners, evidence schema. Domain: payment rules, test cases, acceptance and service outcomes.

Sequence shared assets before wider agency

HORIZON	SHARED INSTITUTIONAL ASSET	DOMAIN RESPONSIBILITY / TRADE-OFF
Prove the workflow	Task identity, approved models, baseline measurement and evidence contract.	Payment owner defines behavior; delivery team measures review effort. Keep scope narrow.
Reuse the assurance	Evaluation runners, policy enforcement and reusable failure cases.	Own domain cases, acceptance and outcomes. Shared services need funded owners.
Delegate bounded actions	Scoped tool adapters, recovery exercises and operational evidence.	Service owner accepts consequence and recovery duty. Expand one authority boundary at a time.